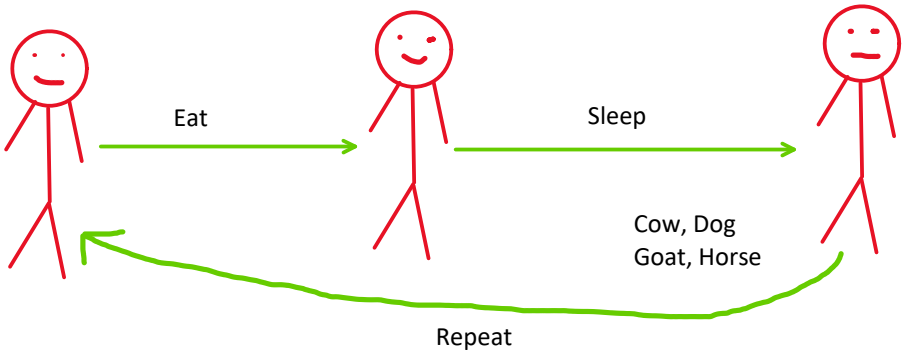
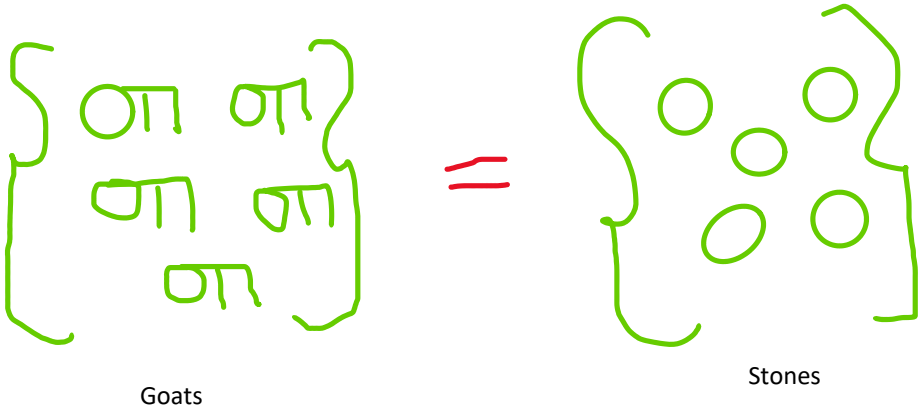
Day - 1 (**Introduction To Programming**)

08 October 2024 22:23

★ 10,000 Years Ago:



- Once upon a time , a man have 5 goats and he went for grazing daily with their goats. At that time they don't know how to speak and how to count. One day he saw that his goats are missing or number of goats are less then previous day.
- At this point they invented the Telly Mark System: 𝍸 𝍸𝍸𝍸 𝍸𝍸 𝍸𝍸𝍸𝍸
- In this system, they keep the stone in pocket and compare the goats with stones.



- **There Are Two Cases:**

  Goats < Stones --> Goat Lost
  Goats = Stones --> No Loss

> ★ Now number of goats are increased then it is difficult to count the goats using telly mark number system.

★ After 500 years later:

New number systems are invented:

- **Egypt:** They invented the Base 60 number system.
- **India:** They invented the Base 10 number system.

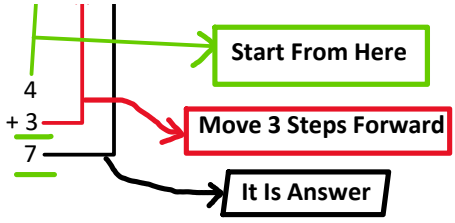
> - Base 10 number system is more popular because it is efficient and it is more familiar to humans, In daily life we saw 10 related things like 10 fingers of hand, 10 finger of feet, etc.
> - Arithmetic Operations are easy in decimal number system.

Base 10 is also called as Decimal Number System

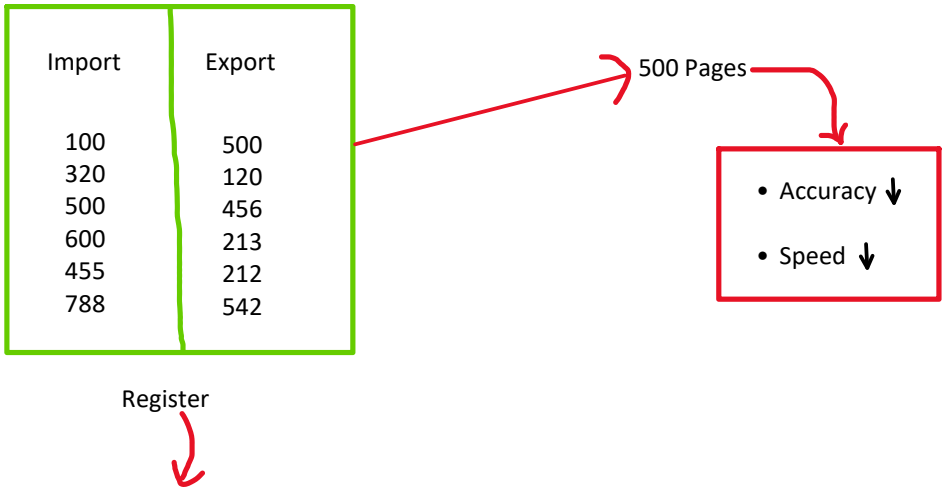
0, 1, 2, 3, 4, 5, 6, 7, 8, 9

4

Start From Here

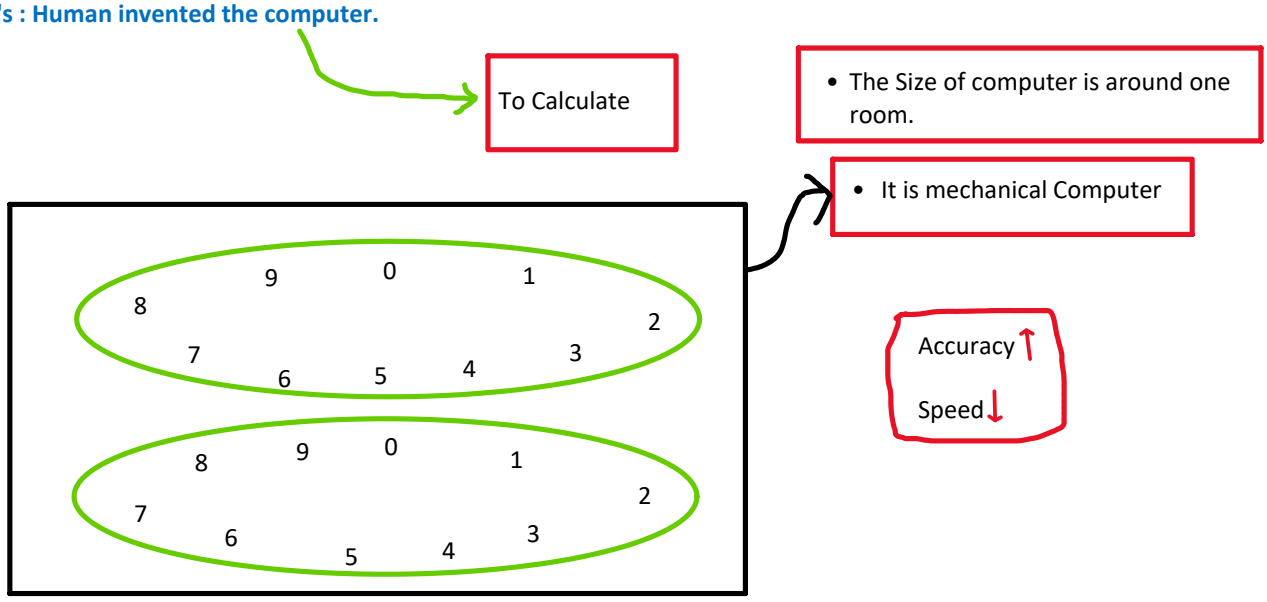


★ **At British Time**, when they import and export the items from one country to another country . At one time the trades are increased and its difficult to calculate the numbers.

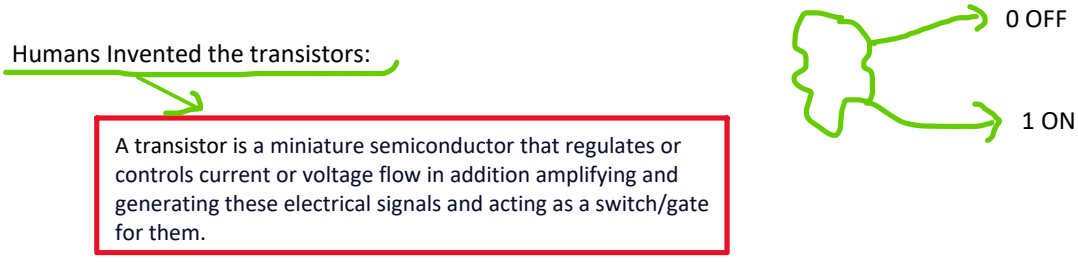


- It is easy to calculate the number of one page but when there are 500 pages then it is difficult to calculate and there is high chance of error, inaccuracy, etc.

★ **In 1800's : Human invented the computer.**

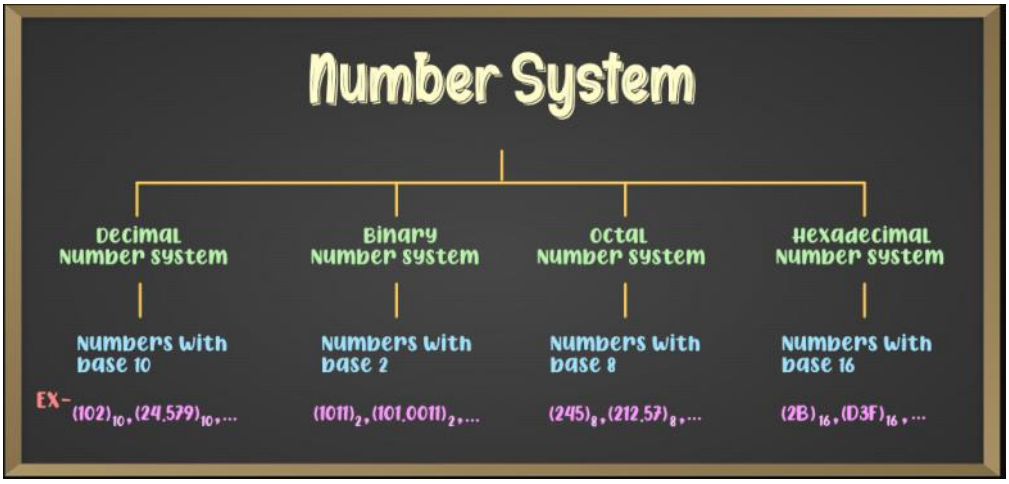


★ **After 100 Years:**



★ Number Systems:

The number system is simply a system to represent or express numbers. There are various types of number systems and the most commonly used ones are decimal number system, binary number system, octal number system, and hexadecimal number system.



- **Binary Number System: (Base 2)** → 0, 1

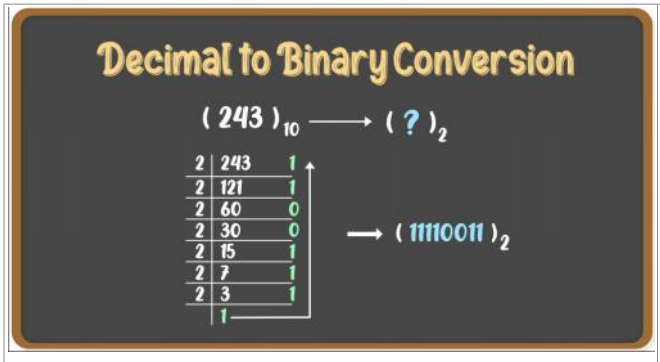
0	0	1	1	10	11
0+	1+	0+	1+	+1	+1
0	1	1	10	11	100

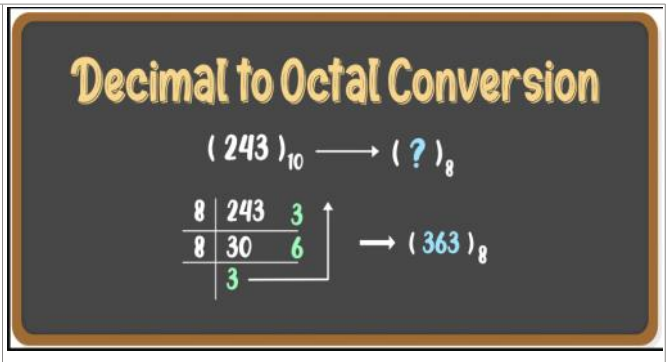
- **Decimal Number System: (Base 10)** → 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- **Octal Number System: (Base 8)** → 0, 1, 2, 3, 4, 5, 6, 7
- **Hexa Number System: (Base 16)** → 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

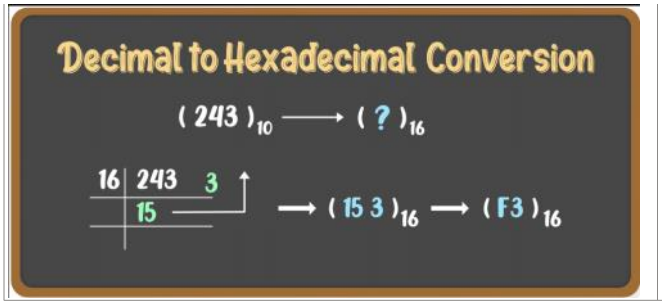
Other Number Systems

★ Conversion Of Number Systems:

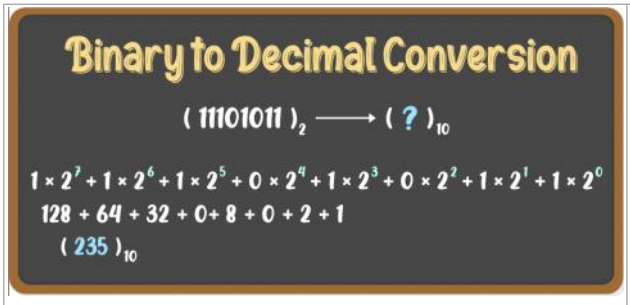
- **Decimal Number System:**

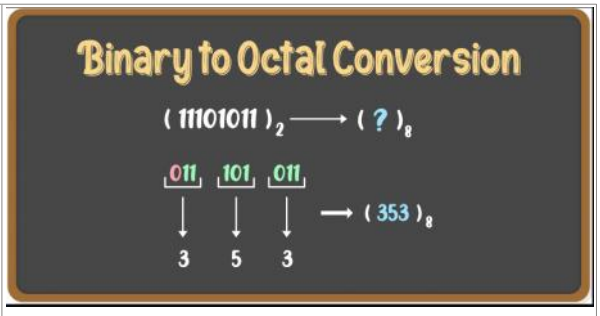


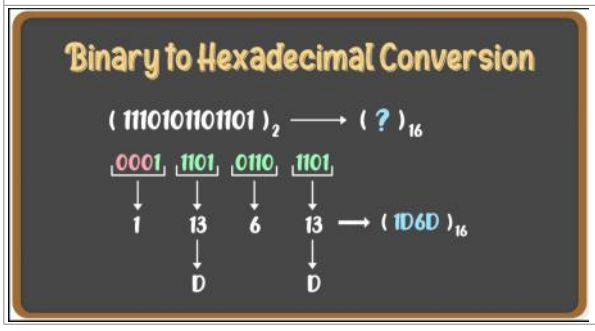




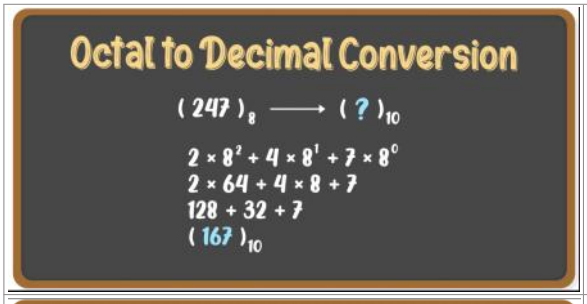
- **Binary Number System:**

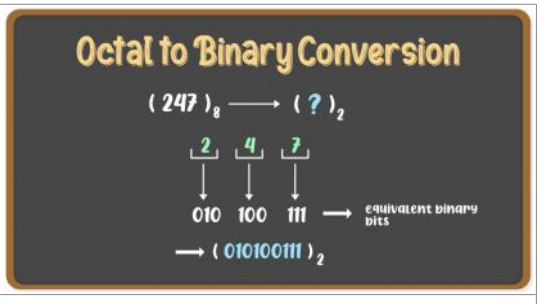


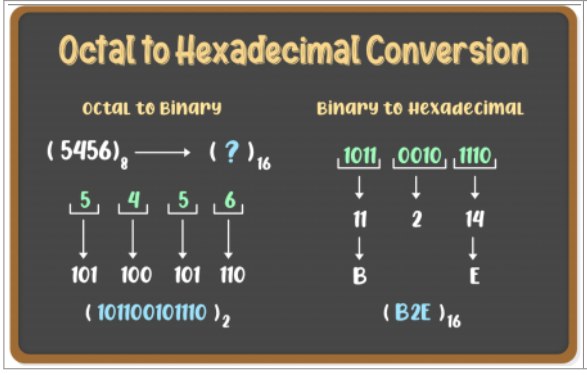




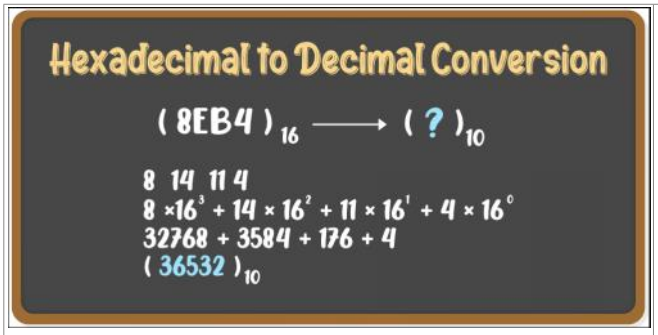
- **Octal Number System:**

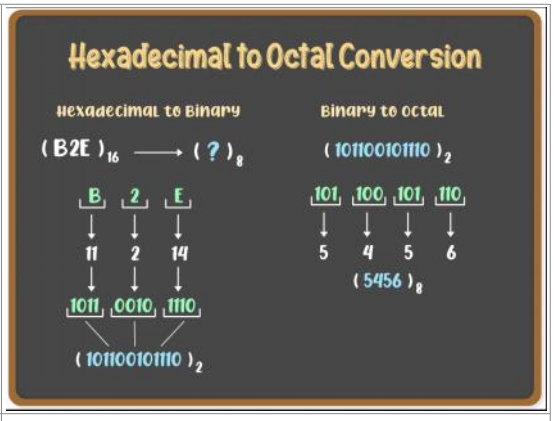


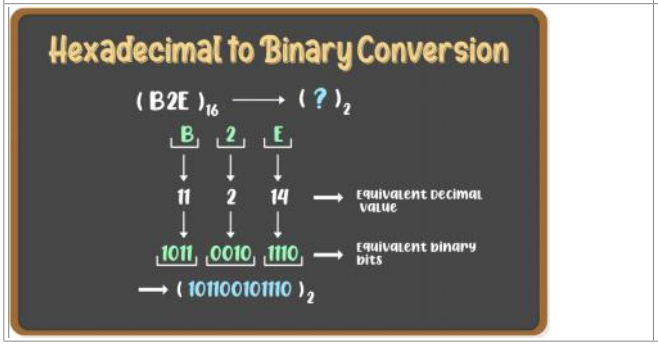




- **Hexa Decimal Number System:**







★ Transistor: (How It Works)

- **There are two different cases:**

Transistor (Decimal)	Transistor (Binary)
0 - OFF 1 - OFF 2 - OFF 3 - ON 4 - OFF 5 - OFF 6 - OFF 7 - ON 8 - OFF 9 - OFF	0 - OFF 0 - OFF 0 - OFF 0 - OFF 0 - OFF 0 - OFF 1 - ON 0 - OFF 1 - ON 0 - OFF
• If we use decimal number system in the transistor, then we can only read the 10 unique numbers in computer. • Example: If you want to read 37 then it can on the bulb of 3 and 7 but here a problem raise where it can also read it as 73 which occurs data lose.	• If we use Binary number system in the transistor, then we can read the 2^{10} unique numbers in computer. • Example: If you want to read 10 then it can on the bulb of 0000001010. It cannot read it as 01. • It is very efficient to use binary number system in the computer.

★ Numbers Store In Transistors:

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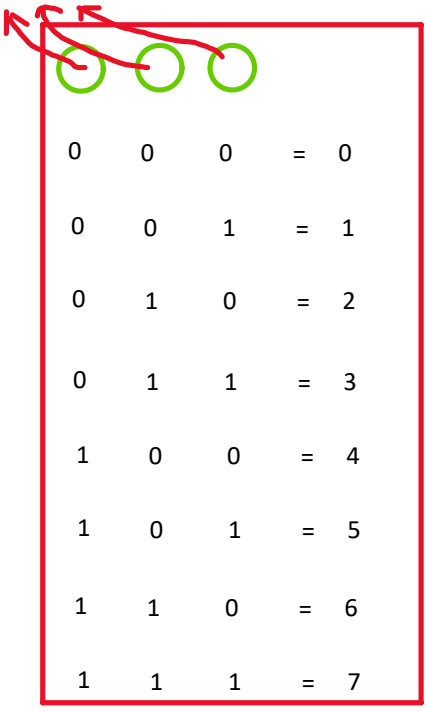
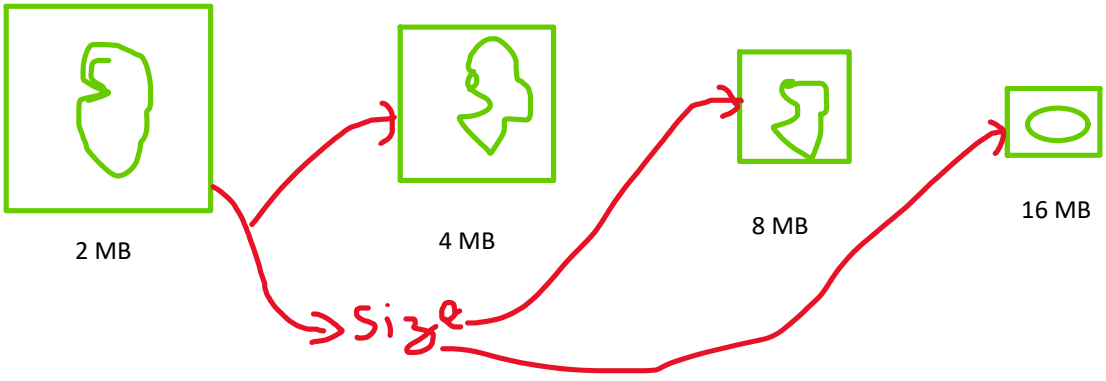


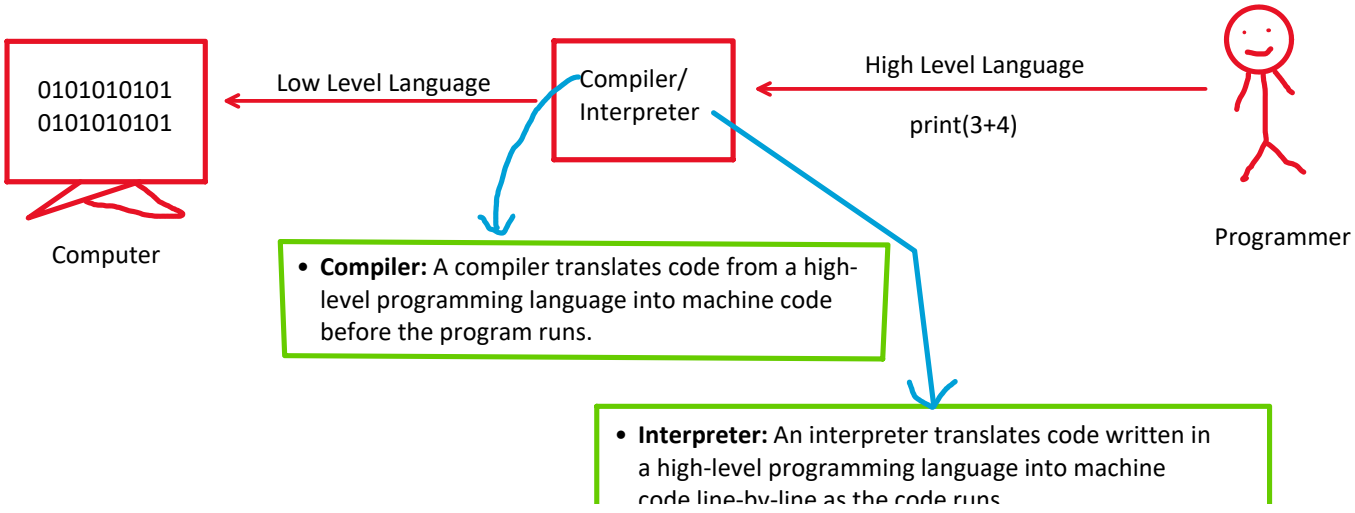
0	0	0	=	0
0	0	1	=	1
0	1	0	=	2
0	1	1	=	3
1	0	0	=	4
1	0	1	=	5
1	1	0	=	6
1	1	1	=	7

★ Moore's Law:

The capacity of the transistor is increased and the size of transistor is decreased in every two years.



★ Computer:

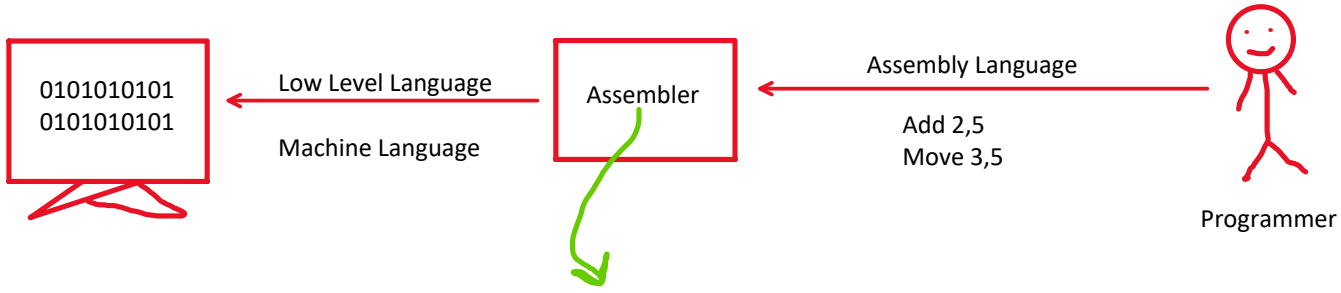


- **Interpreter:** An interpreter translates code written in a high-level programming language into machine code line-by-line as the code runs.

★ Speed Of Programming Language:

Machine Language >> Assembly Language >> High Level Language

★ Assembler:



Assembler is a program for converting instructions written in low-level assembly code into relocatable machine code and generating along information for the loader. It is necessary to convert user-written programs into machinery code. This is called a translation of a high-level language to a low-level that is machinery language.

★ 10 Years Ago:

- Humans can use Pendrive, CD, Floopy Disk to transfer the data.
- When internet comes then the humans transfer the data over the internet and it increase the data to store on the internet.
- At this time we need data structure to store data in efficient way.

★ Data:

Store And Fetch

Less Time
Less Space

Data Structure: Store Data In Efficient Way, When We Fetch The Data It Takes Less Time And Less Space.

Algorithms: It helps to execute code very fast.

Time Complexity: The time complexity of an algorithm quantifies the amount of time taken by an algorithm to run as a function of the length of the input.

Space Complexity: Problem-solving using computer requires memory to hold temporary data or final result

Time Complexity: The time complexity of an algorithm quantifies the amount of time taken by an algorithm to run as a function of the length of the input.

Space Complexity: Problem-solving using computer requires memory to hold temporary data or final result while the program is in execution. The amount of memory required by the algorithm to solve given problem is called space complexity of the algorithm.

★ **Example:**

We gave one question to both: The question is sum of natural number then gudu bhaiya do it by add one by one and bablu bhaiya use sum of natural number formula.

`

Gudu Bhaiya

1+2+3+4+5+6+7+8+9+10+11 ……..500 numbers

Bablu Bhaiya

n(n+1)/2

Bablu Bhaiya is more fast and efficient than Gudu bhaiya.